

Data Science & Analytics – Task 1

Business Sales Performance Analytics

Internship Project Documentation

Prepared using: Microsoft Power BI

Dataset: Superstore Sales Dataset

Project Type: Business Sales Performance Analytics

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1. Executive Summary

The **Business Sales Performance Analytics** project was developed as part of the **Future Interns Data Science & Analytics – Task 1**.

The objective of this project was to transform raw business sales data into a **professional, client-ready Power BI dashboard** that helps stakeholders understand revenue, profitability, product performance, category performance, regional performance, and customer behavior.

The project follows a complete analytics workflow:

Raw Sales Data → Data Cleaning → Data Preparation → KPI Analysis → Trend Analysis → Performance Analysis → Insights → Recommendations

The final Power BI solution consists of **three interactive dashboard pages**:

1. **Sales Performance Dashboard**
2. **Product Performance & Profitability**
3. **Regional & Customer Insights**

The dashboard enables users to analyze business performance through interactive filters and visualizations and provides actionable insights that can support business decision-making.

2. About The Project

Future Interns Task 1 focuses on performing **Business Sales Performance Analytics** using real-world sales data.

The task requires the analyst to:

- Clean and organize raw sales data
- Analyze revenue trends
- Identify top-selling products
- Analyze high-value categories and regions
- Build a professional dashboard or analysis report
- Generate clear business insights
- Provide actionable recommendations
- Document the project properly through GitHub and a dashboard/project link

This project addresses these requirements using **Microsoft Power BI**.

3. Business Problem

Businesses generate large amounts of sales data, but raw transactional records do not immediately provide useful business information.

A business stakeholder needs answers to questions such as:

- Which products generate the most revenue?
- How do sales change over time?
- Which categories perform best?
- Which regions are most profitable?
- Which products generate losses?
- Which customer segments contribute the most revenue?

- Are discounts affecting profitability?
- Where should the business focus to improve performance?

The purpose of this project is to answer these questions through an interactive analytical dashboard.

4. Project Objectives

The project objectives are directly aligned with the Future Interns Task 1 requirements.

Primary Objectives

1. Clean and organize raw sales data.
 2. Analyze revenue performance over time.
 3. Identify top-selling products.
 4. Analyze category performance.
 5. Analyze regional performance.
 6. Evaluate profitability.
 7. Develop meaningful business KPIs.
 8. Identify high- and low-performing products.
 9. Analyze customer and segment performance.
 10. Generate actionable business insights.
 11. Present findings through a professional, client-ready dashboard.
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5. Dataset Description

Dataset Used

Superstore Sales Dataset

The dataset represents a retail business scenario containing transactional information about customer orders, products, regions, sales, discounts, and profit.

Main Information Available

- Order information
- Customer information
- Product information
- Category and sub-category
- Geographic information
- Sales
- Quantity
- Discount
- Profit
- Shipping information

The dataset is suitable for analyzing the business questions specified in the internship task.

6. Data Cleaning & Preparation

Data cleaning and preparation were performed using **Power Query in Power BI**.

Data Preparation Activities

6.1 Data Type Validation

Important fields were assigned appropriate data types.

Field	Data Type
Order Date	Date
Ship Date	Date
Sales	Decimal
Profit	Decimal
Quantity	Whole Number
Discount	Decimal

6.2 Data Quality Review

The dataset was reviewed for:

- Missing values
- Duplicate records
- Incorrect data types
- Inconsistent values
- Numerical field validity

6.3 Data Preparation

The cleaned data was then loaded into Power BI for analysis.

Result

The prepared dataset was suitable for creating KPIs, trend analysis, product analysis, regional analysis, and interactive dashboard visualizations.

7. Business Kpi Analysis

The dashboard uses key performance indicators to provide an immediate overview of business performance.

KPI 1 — Total Sales

2.30M

Represents the total revenue generated from all orders.

KPI 2 — Total Profit

286.40K

Represents the overall profit generated.

KPI 3 — Total Orders

5,009

Represents the number of unique orders.

KPI 4 — Total Quantity

38K

Represents the total number of units sold.

KPI 5 — Average Order Value

\$459

Calculated as:

Total Sales ÷ Total Orders

It represents the average revenue generated per order.

KPI 6 — Profit Margin

12.47%

Calculated as:

$\text{Total Profit} \div \text{Total Sales} \times 100$

It indicates the percentage of sales retained as profit.

8. Dashboard Design & Analysis

The dashboard was designed as a **three-page analytical report**.

The pages follow a logical business storytelling structure:

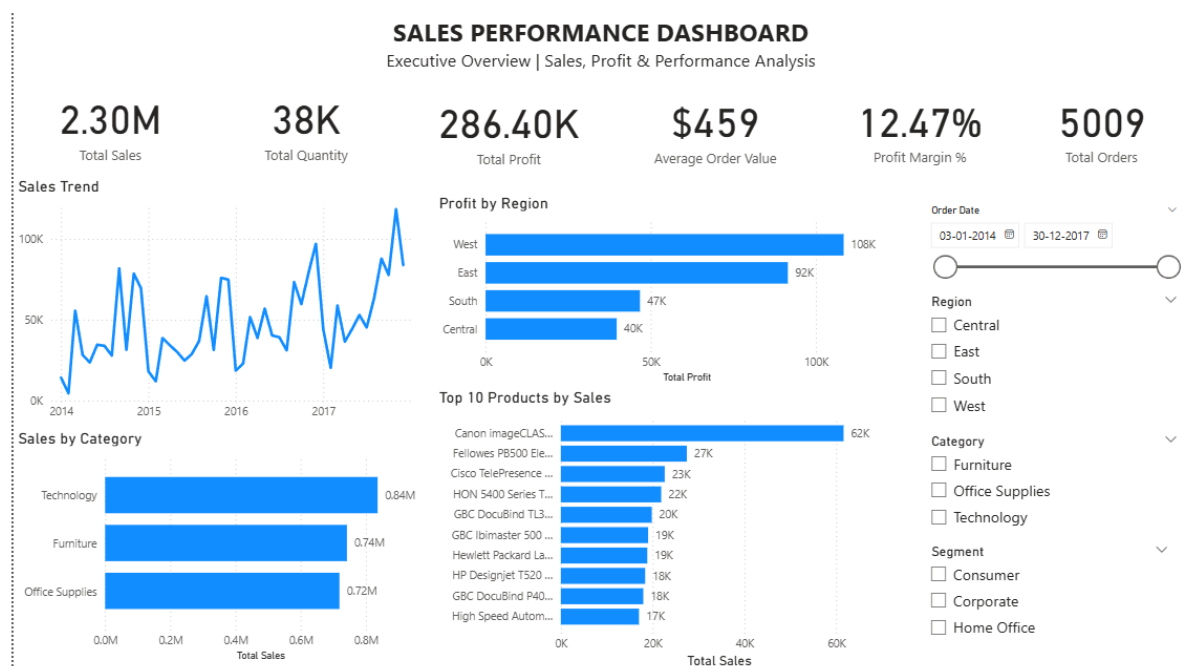
Overall Performance → Product Profitability → Regional & Customer Analysis

9. Page 1 — Sales Performance Dashboard

Purpose

The first page provides an executive-level overview of business sales performance.

Visualizations



Sales Trend

Shows how sales change over time.

This allows stakeholders to identify:

- Growth patterns
- Declining periods
- Sales fluctuations
- Overall trends

Sales by Category

Compares sales across:

- Technology
- Furniture
- Office Supplies

Profit by Region

Compares profitability between:

- West
- East
- South
- Central

Top 10 Products by Sales

Identifies the products generating the highest revenue.

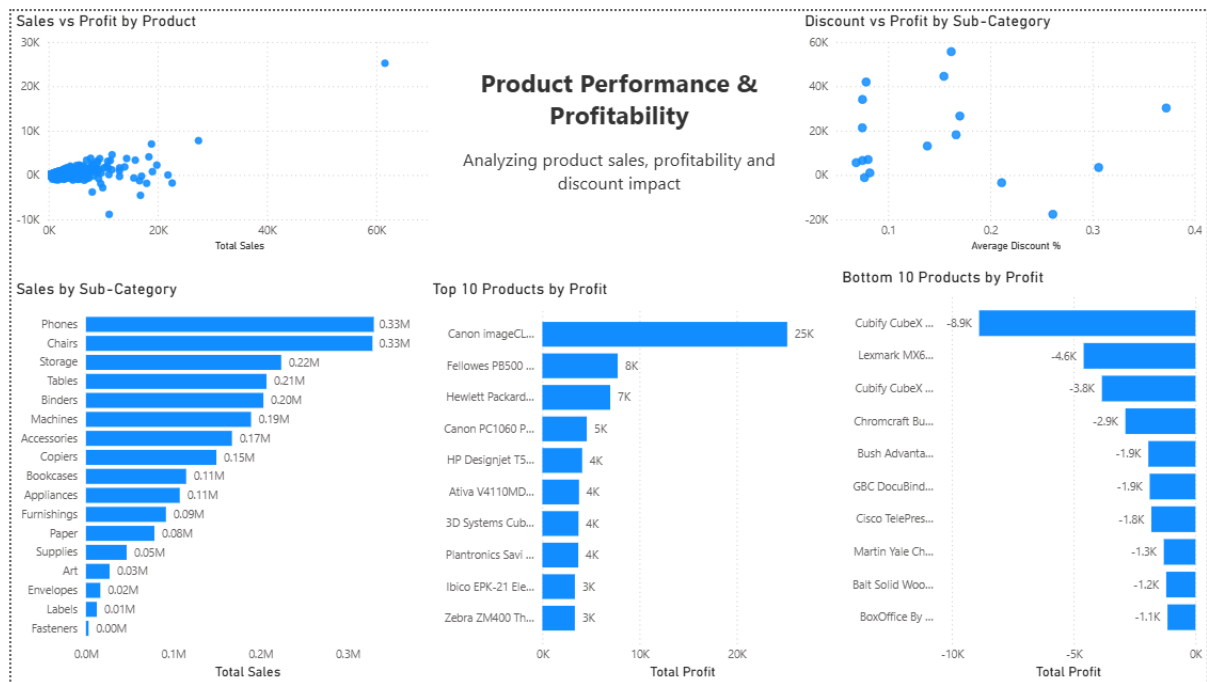
Interactive Filters

- Order Date
- Region
- Category
- Segment

10. Page 2 — Product Performance & Profitability

Purpose

This page focuses on product-level analysis and profitability.



Sales vs Profit by Product

Compares sales and profit across products.

This helps identify products that have:

- High sales and high profit
- High sales but low profit
- Negative profit

Sales by Sub-Category

Shows sales performance across product sub-categories.

Average Discount vs Profit

Analyzes the relationship between discounting and profitability.

Top 10 Products by Profit

Identifies the most profitable products.

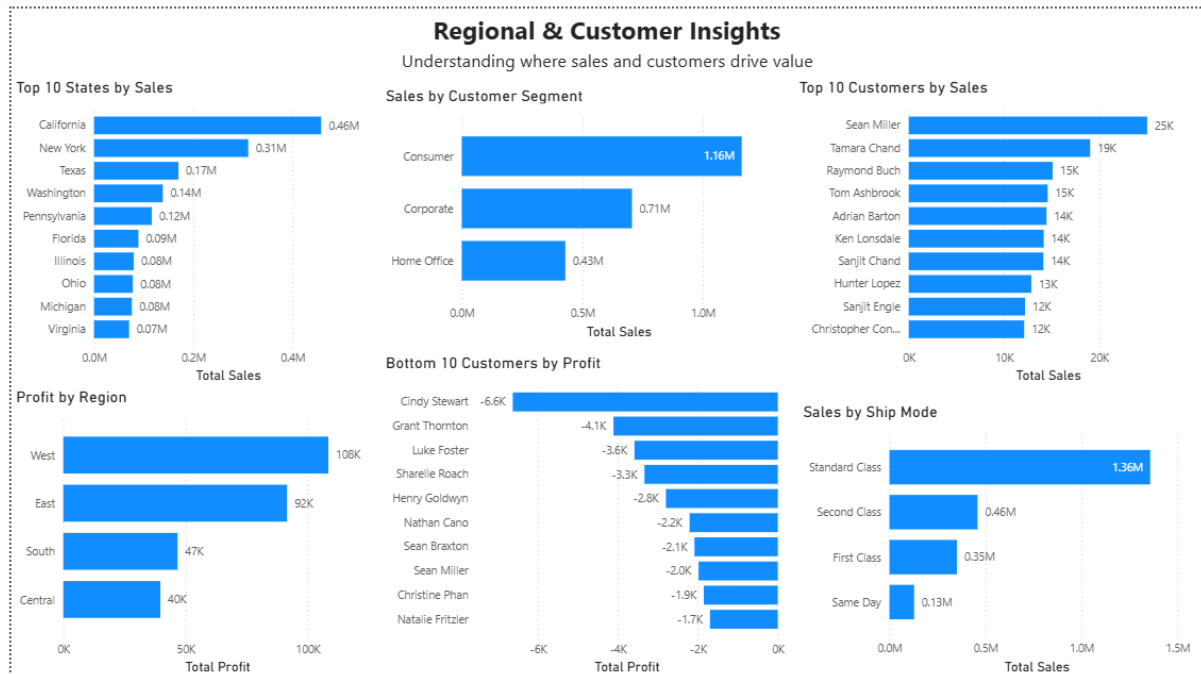
Bottom 10 Products by Profit

Identifies products generating the lowest or negative profit.

11. Page 3 — Regional & Customer Insights

Purpose

This page provides geographic, customer, and operational analysis.



Top 10 States by Sales

Identifies the strongest states by revenue.

Sales by Customer Segment

Compares:

- Consumer
- Corporate
- Home Office

Top 10 Customers by Sales

Identifies high-value customers.

Bottom 10 Customers by Profit

Identifies customers associated with low profitability.

Profit by Region

Provides a regional profitability comparison.

Sales by Ship Mode

Analyzes sales across different shipping methods.

12. Revenue Trend Analysis

The Sales Trend visualization provides a time-based view of revenue performance.

The analysis allows stakeholders to understand how revenue changes across the available period and identify periods of stronger or weaker performance.

This directly addresses the internship requirement:

"How do sales change over time?"

The interactive date filter also allows users to focus on specific periods.

13. Product Performance Analysis

Product performance was analyzed from two perspectives:

Revenue

Using:

Top 10 Products by Sales

to identify products generating the highest revenue.

Profitability

Using:

Top 10 Products by Profit

and

Bottom 10 Products by Profit

to distinguish between products that generate revenue and products that actually generate profit.

This is important because:

A product generating high sales does not necessarily generate high profit.

14. Category & Regional Performance

Category Analysis

The dashboard shows:

Category	Approx. Sales
Technology	0.84M
Furniture	0.74M
Office Supplies	0.72M

Technology is the highest-selling category.

Regional Analysis

Approximate regional profit:

Region	Profit
West	108K
East	92K
South	47K
Central	40K

West is the strongest region in terms of profit.

This analysis directly addresses the internship requirement to identify **high-value regions**.

15. Customer & Operational Analysis

Customer Segment

Approximate sales:

Segment	Sales
Consumer	1.16M
Corporate	0.71M

Segment	Sales
Home Office	0.43M

Consumer is the largest sales-generating segment.

Shipping

Standard Class is the dominant shipping method by sales, generating approximately **1.36M**.

This provides an additional operational perspective beyond the core sales analysis.

16. Key Business Insights

1. Technology leads sales

Technology generates approximately **0.84M**, making it the highest-selling category.

Meaning: Technology represents an important revenue-generating area.

2. West is the strongest region by profit

West generates approximately **108K profit**, significantly higher than South and Central.

Meaning: Regional performance varies considerably and should be investigated.

3. Consumer is the largest segment

Consumer contributes approximately **1.16M sales**.

Meaning: Consumer customers represent the largest revenue opportunity.

4. Standard Class dominates shipping

Standard Class contributes approximately **1.36M sales**.

Meaning: It is the primary shipping method used within the dataset.

5. Product revenue and profitability are different

Some products generate significant sales but relatively low or negative profit.

Meaning: Product performance should be evaluated using both revenue and profitability

6. Discounting requires monitoring

The discount-profit analysis indicates that some highly discounted sub-categories have weak profitability.

Meaning: Excessive discounting can potentially reduce margins.

17. Actionable Business Recommendations

1.Review loss-making products

Management should investigate the Bottom 10 Products by Profit.

Review:

- Pricing
- Costs
- Discounts
- Demand
- Profit margin

2. Optimize discount strategy

Discounts should be aligned with profitability.

Instead of applying aggressive discounts broadly, the business should identify where discounts actually generate profitable additional sales.

3.Strengthen high-performing categories

Technology is the leading category by sales.

The business should focus on:

- Maintaining stock availability
- Promoting successful products
- Identifying high-margin Technology products

4.Investigate weaker regions

South and Central generate substantially less profit than West.

Management should investigate differences in:

- Product mix
- Pricing
- Discounts
- Customer composition
- Sales volume

5.Retain high-value customers

Top customers by sales should be identified for targeted customer-retention strategies.

6.Evaluate profitability, not just revenue

Products and customers should be evaluated using both sales and profit to avoid focusing solely on high-revenue but low-margin business.

18. Business Impact

The dashboard provides a centralized view of business performance and can help stakeholders:

- Monitor sales performance
- Track profitability
- Identify top-selling products
- Identify loss-making products
- Compare regional performance
- Understand customer segments
- Evaluate discounting
- Support business decision-making

The dashboard transforms raw transaction-level data into a format that can be quickly understood by business owners, managers, and other stakeholders.

19. Skills & Learning Outcomes

The project directly demonstrates the skills mentioned in the internship task.

Data Cleaning & Preparation

Used Power Query to prepare and organize raw sales data.

Business-Focused KPI Analysis

Created KPIs including:

- Sales
- Profit
- Orders
- Quantity
- AOV
- Profit Margin

Trend & Performance Analysis

Analyzed:

- Sales trends
- Product performance
- Category performance
- Regional performance
- Customer performance

Insight Generation

Converted visual patterns into business observations.

Business Storytelling

Structured the dashboard into three pages to communicate information progressively:

Executive Overview → Product Analysis → Regional & Customer Analysis

20. Conclusion

The **Business Sales Performance Analytics Dashboard** successfully addresses the objectives of **Future Interns Data Science & Analytics – Task 1**.

The project demonstrates the complete process of:

Cleaning → Analyzing → Visualizing → Interpreting → Recommending

The final dashboard provides stakeholders with an interactive view of sales, revenue trends, products, categories, regions, customers, profitability, and discounts.

The analysis highlights important business opportunities such as strengthening high-performing categories, reviewing loss-making products, optimizing discounts, and investigating regional profitability differences.

The project therefore demonstrates practical application of **Power BI and business analytics** to a real-world sales scenario.

21. Future Scope

Although the current dashboard satisfies the core Task 1 objectives, future improvements could include:

- Sales forecasting
- Profit forecasting
- Customer segmentation
- Customer lifetime value
- What-if discount analysis
- Automated data refresh
- Advanced drill-through analysis
- Predictive analytics

These features could extend the project from descriptive analytics toward predictive and prescriptive analytics.